1. Arrays & Hashing

#	Question	Pattern	Priority
1	Two Sum	HashMap lookup	P0
2	Contains Duplicate	Set duplicate detection	P0
3	Valid Anagram	Frequency map	P0
4	Group Anagrams	Hash key grouping	P0
5	Top K Frequent Elements	Frequency + heap/bucket	P0
6	Product of Array Except Self	Prefix/suffix	P0
7	Longest Consecutive Sequence	Set sequence expansion	P0
8	Encode and Decode Strings	String design/parsing	P0
9	Subarray Sum Equals K	Prefix sum + HashMap	P0
10	Majority Element	Counting / Boyer-Moore	P1
11	Intersection of Two Arrays	Set operations	P1
12	Intersection of Two Arrays II	Frequency map	P1
13	Find All Numbers Disappeared in an Array	Index marking	P1
14	First Missing Positive	Index placement	P1
15	Sort Characters By Frequency	Frequency sorting	P1
16	Ransom Note	Frequency map	P1
17	Isomorphic Strings	Two-way mapping	P1
18	Word Pattern	HashMap mapping	P1
19	Design HashMap	Hash table design	P2
20	Insert Delete GetRandom O(1)	HashMap + array	P0

2. Strings

#	Question	Pattern	Priority
1	Valid Palindrome	Two pointers	P0

#	Question	Pattern	Priority
2	Longest Common Prefix	String scanning	P1
3	Reverse Words in a String	String manipulation	P1
4	String to Integer Atoi	Parsing edge cases	P1
5	Implement strStr / Find Index of First Occurrence	Substring search	P1
6	Longest Palindromic Substring	Expand around center	P0
7	Palindromic Substrings	Expand around center	P1
8	Valid Parenthesis String	Greedy/string	P1
9	Minimum Remove to Make Valid Parentheses	Stack/string cleanup	P1
10	Decode String	Stack/string parsing	P0
11	Basic Calculator	Stack/parsing	P1
12	Basic Calculator II	Stack/parsing	P1
13	Compare Version Numbers	String split/two pointers	P1
14	Multiply Strings	Simulation	P1
15	Zigzag Conversion	Simulation	P2
16	Roman to Integer	Map parsing	P1
17	Integer to Roman	Greedy mapping	P2
18	Add Binary	String addition	P1
19	Simplify Path	Stack/string	P1
20	Text Justification	Simulation	P2

3. Two Pointers

#	Question	Pattern	Priority
1	Two Sum II - Input Array Is Sorted	Opposite pointers	P0
2	3Sum	Sort + two pointers	P0
3	4Sum	Sort + two pointers	P1
4	Container With Most Water	Opposite pointers	P0

#	Question	Pattern	Priority
5	Trapping Rain Water	Two pointers	P0
6	Move Zeroes	Slow/fast pointer	P0
7	Remove Duplicates from Sorted Array	Slow/fast pointer	P0
8	Remove Element	Slow pointer	P1
9	Sort Colors	Dutch National Flag	P0
10	Squares of a Sorted Array	Two pointers	P1
11	Backspace String Compare	Reverse pointers/stack	P1
12	Merge Sorted Array	Reverse merge	P1
13	Valid Palindrome II	Two pointers with skip	P1
14	Partition Labels	Greedy + last index	P1
15	Boats to Save People	Greedy two pointers	P1
16	Linked List Cycle	Slow/fast pointer	P0
17	Middle of the Linked List	Slow/fast pointer	P1
18	Palindrome Linked List	Slow/fast + reverse	P1
19	Remove Nth Node From End	Gap pointer	P0
20	Find the Duplicate Number	Floyd cycle	P1

4. Sliding Window

#	Question	Pattern	Priority
1	Best Time to Buy and Sell Stock	Running min/window	P0
2	Longest Substring Without Repeating Characters	Dynamic window + set	P0
3	Longest Repeating Character Replacement	Dynamic window + frequency	P0
4	Permutation in String	Fixed window + frequency	P0
5	Minimum Window Substring	Dynamic window + map	P0
6	Sliding Window Maximum	Deque	P0

#	Question	Pattern	Priority
7	Find All Anagrams in a String	Fixed window + frequency	P0
8	Maximum Average Subarray I	Fixed window	P1
9	Minimum Size Subarray Sum	Dynamic window	P1
10	Fruit Into Baskets	At most K types	P1
11	Longest Substring with At Most K Distinct Characters	Dynamic window	P1
12	Max Consecutive Ones III	Dynamic window	P1
13	Subarrays with K Different Integers	At most K technique	P1
14	Count Number of Nice Subarrays	Prefix/window	P1
15	Frequency of the Most Frequent Element	Sort + window	P1
16	Minimum Number of Flips to Make Binary String Alternating	Sliding window	P2
17	Grumpy Bookstore Owner	Fixed window gain	P1
18	Repeated DNA Sequences	Rolling window/hash	P1
19	Minimum Window Subsequence	Window/DP	P2
20	Longest Turbulent Subarray	Window state	P2

5. Prefix Sum & Kadane

#	Question	Pattern	Priority
1	Range Sum Query Immutable	Prefix sum	P0
2	Subarray Sum Equals K	Prefix + HashMap	P0
3	Maximum Subarray	Kadane	P0
4	Product of Array Except Self	Prefix/suffix product	P0
5	Find Pivot Index	Prefix/suffix sum	P0

#	Question	Pattern	Priority
6	Continuous Subarray Sum	Prefix modulo	P1
7	Subarray Sums Divisible by K	Prefix modulo count	P1
8	Maximum Product Subarray	Prefix-style DP	P1
9	Minimum Size Subarray Sum	Prefix/window	P1
10	Corporate Flight Bookings	Difference array	P1
11	Car Pooling	Difference array	P1
12	Range Addition	Difference array	P1
13	Number of Sub-arrays of Size K and Average Greater than Threshold	Fixed window prefix	P1
14	Matrix Block Sum	2D prefix	P1
15	Range Sum Query 2D Immutable	2D prefix	P1
16	Count Vowels Permutation	DP prefix-like	P2
17	Maximum Sum Circular Subarray	Kadane variant	P1
18	Contiguous Array	Prefix balance + HashMap	P1
19	Binary Subarrays With Sum	Prefix count/window	P1
20	Maximum Size Subarray Sum Equals k	Prefix + HashMap	P1

6. Binary Search

#	Question	Pattern	Priority
1	Binary Search	Classic binary search	P0
2	Search Insert Position	Lower bound	P0
3	Search in Rotated Sorted Array	Modified binary search	P0
4	Find Minimum in Rotated Sorted Array	Modified binary search	P0
5	Search a 2D Matrix	Binary search matrix	P0
6	Koko Eating Bananas	Binary search answer	P0

#	Question	Pattern	Priority
7	Find Peak Element	Binary decision	P0
8	Median of Two Sorted Arrays	Partition binary search	P1
9	Search in Rotated Sorted Array II	Duplicates variant	P1
10	Find First and Last Position of Element in Sorted Array	Lower/upper bound	P1
11	Sqrt(x)	Binary search answer	P1
12	Capacity To Ship Packages Within D Days	Binary search answer	P1
13	Split Array Largest Sum	Binary search answer	P1
14	Minimize Maximum Distance to Gas Station	Binary search answer	P2
15	Time Based Key-Value Store	HashMap + binary search	P0
16	Find K Closest Elements	Binary search + window	P1
17	Find Smallest Letter Greater Than Target	Binary search	P1
18	Peak Index in a Mountain Array	Binary decision	P1
19	Single Element in a Sorted Array	Parity binary search	P1
20	Find Minimum in Rotated Sorted Array II	Duplicates variant	P2

7. Stack & Queue

#	Question	Pattern	Priority
1	Valid Parentheses	Stack	P0
2	Min Stack	Stack design	P0
3	Evaluate Reverse Polish Notation	Stack evaluation	P0
4	Generate Parentheses	Stack/backtracking	P0
5	Daily Temperatures	Monotonic stack	P0
6	Car Fleet	Stack after sorting	P0
7	Largest Rectangle in	Monotonic stack	P0

#	Question	Pattern	Priority
	Histogram		
8	Decode String	Stack parsing	P0
9	Simplify Path	Stack	P1
10	Remove All Adjacent Duplicates in String	Stack	P1
11	Asteroid Collision	Stack simulation	P1
12	Online Stock Span	Monotonic stack	P1
13	Next Greater Element I	Monotonic stack	P1
14	Next Greater Element II	Circular monotonic stack	P1
15	132 Pattern	Monotonic stack	P2
16	Implement Queue using Stacks	Stack design	P1
17	Implement Stack using Queues	Queue design	P1
18	Design Circular Queue	Queue design	P1
19	Moving Average from Data Stream	Queue	P1
20	First Unique Character in a Stream	Queue + map	P2

8. Linked List

#	Question	Pattern	Priority
1	Reverse Linked List	Pointer reversal	P0
2	Merge Two Sorted Lists	Merge pointers	P0
3	Linked List Cycle	Slow/fast pointer	P0
4	Reorder List	Middle + reverse + merge	P0
5	Remove Nth Node From End	Two pointers	P0
6	Copy List with Random Pointer	HashMap/deep copy	P0
7	Add Two Numbers	Simulation	P0
8	Merge K Sorted Lists	Heap/merge	P0
9	Palindrome Linked List	Reverse second half	P1
10	Intersection of Two	Pointer switching	P1

#	Question	Pattern	Priority
	Linked Lists		
11	Middle of the Linked List	Slow/fast	P1
12	Swap Nodes in Pairs	Pointer manipulation	P1
13	Reverse Nodes in k-Group	Advanced reversal	P1
14	Rotate List	Length + pointer	P1
15	Partition List	Two dummy lists	P1
16	Sort List	Merge sort linked list	P1
17	Remove Duplicates from Sorted List	Pointer cleanup	P1
18	Remove Duplicates from Sorted List II	Pointer cleanup	P1
19	Flatten a Multilevel Doubly Linked List	DFS/list	P2
20	LRU Cache	HashMap + doubly linked list	P0

9. Trees & BST

#	Question	Pattern	Priority
1	Maximum Depth of Binary Tree	DFS	P0
2	Same Tree	DFS compare	P0
3	Invert Binary Tree	DFS	P0
4	Binary Tree Level Order Traversal	BFS	P0
5	Validate Binary Search Tree	Bounds DFS	P0
6	Kth Smallest Element in BST	Inorder traversal	P0
7	Lowest Common Ancestor of BST	BST traversal	P0
8	Construct Binary Tree from Preorder and Inorder Traversal	Recursion + map	P0
9	Binary Tree Maximum Path Sum	DFS gain	P0
10	Serialize and Deserialize Binary Tree	Tree encoding/design	P0

#	Question	Pattern	Priority
11	Subtree of Another Tree	DFS compare	P1
12	Diameter of Binary Tree	DFS height	P1
13	Balanced Binary Tree	DFS height	P1
14	Path Sum	DFS	P1
15	Path Sum II	DFS backtracking	P1
16	Right Side View	BFS/DFS	P1
17	Count Good Nodes in Binary Tree	DFS with max	P1
18	Lowest Common Ancestor of Binary Tree	Recursive DFS	P1
19	Populating Next Right Pointers in Each Node	BFS/pointer	P1
20	Recover Binary Search Tree	Inorder anomaly	P2

10. Heap / Priority Queue

#	Question	Pattern	Priority
1	Kth Largest Element in an Array	Heap/quickselect	P0
2	Top K Frequent Elements	Heap/bucket	P0
3	Find Median from Data Stream	Two heaps	P0
4	Merge K Sorted Lists	Min heap	P0
5	Task Scheduler	Heap + greedy	P0
6	K Closest Points to Origin	Heap/quickselect	P0
7	Meeting Rooms II	Min heap	P0
8	Last Stone Weight	Max heap	P1
9	Kth Largest Element in a Stream	Min heap	P1
10	Reorganize String	Max heap	P1
11	Minimum Cost to Connect Sticks	Min heap	P1
12	Ugly Number II	Min heap/DP	P1

#	Question	Pattern	Priority
13	Find K Pairs with Smallest Sums	Min heap	P1
14	IPO	Two heaps/greedy	P2
15	The Skyline Problem	Heap/sweep line	P2
16	Smallest Range Covering Elements from K Lists	Heap + window	P2
17	Seat Reservation Manager	Min heap	P1
18	Single-Threaded CPU	Heap scheduling	P1
19	Process Tasks Using Servers	Two heaps	P2
20	Design Twitter	Heap + maps	P1

11. Graphs & Matrix BFS/DFS

#	Question	Pattern	Priority
1	Number of Islands	Grid DFS/BFS	P0
2	Clone Graph	Graph copy + map	P0
3	Max Area of Island	Grid DFS	P0
4	Pacific Atlantic Water Flow	Reverse DFS/BFS	P0
5	Rotting Oranges	Multi-source BFS	P0
6	Course Schedule	Directed cycle	P0
7	Course Schedule II	Topological sort	P0
8	Graph Valid Tree	DFS/Union Find	P0
9	Number of Connected Components in an Undirected Graph	DFS/Union Find	P0
10	Word Ladder	BFS shortest path	P0
11	Walls and Gates	Multi-source BFS	P1
12	Surrounded Regions	Boundary DFS	P1
13	Shortest Path in Binary Matrix	BFS	P1
14	01 Matrix	Multi-source BFS	P1
15	Redundant Connection	Union Find	P1
16	Accounts Merge	Union Find/DFS	P1
17	Alien Dictionary	Topological sort	P1

#	Question	Pattern	Priority
18	Network Delay Time	Dijkstra	P1
19	Cheapest Flights Within K Stops	Graph shortest path variant	P1
20	Reconstruct Itinerary	DFS/Euler path	P2

12. Intervals

#	Question	Pattern	Priority
1	Merge Intervals	Sort + merge	P0
2	Insert Interval	Linear merge	P0
3	Non-overlapping Intervals	Greedy	P0
4	Meeting Rooms	Sort intervals	P0
5	Meeting Rooms II	Min heap/sweep line	P0
6	Employee Free Time	Merge intervals	P1
7	Minimum Number of Arrows to Burst Balloons	Greedy intervals	P1
8	Interval List Intersections	Two pointers	P1
9	Remove Covered Intervals	Sort + scan	P1
10	My Calendar I	Interval conflict	P1
11	My Calendar II	Sweep line	P2
12	My Calendar III	Sweep line/tree map	P2
13	Car Pooling	Difference array	P1
14	Corporate Flight Bookings	Difference array	P1
15	Summary Ranges	Interval construction	P1
16	Data Stream as Disjoint Intervals	Ordered intervals	P2
17	Partition Labels	Greedy intervals	P1
18	Queue Reconstruction by Height	Sorting/greedy	P1
19	Maximum Number of Events That Can Be Attended	Greedy + heap	P1
20	Range Module	Ordered intervals	P2

13. Backtracking

#	Question	Pattern	Priority
1	Subsets	Include/exclude	P0
2	Permutations	Swap/used array	P0
3	Combination Sum	Backtracking	P0
4	Letter Combinations of a Phone Number	Backtracking	P0
5	Word Search	Grid backtracking	P0
6	Generate Parentheses	Backtracking	P0
7	Combination Sum II	Backtracking with duplicates	P1
8	Combinations	Backtracking	P1
9	Subsets II	Duplicates handling	P1
10	Permutations II	Duplicates handling	P1
11	Palindrome Partitioning	Backtracking	P1
12	N-Queens	Backtracking	P1
13	Sudoku Solver	Constraint backtracking	P2
14	Restore IP Addresses	Backtracking	P1
15	Expression Add Operators	Backtracking	P2
16	Partition to K Equal Sum Subsets	Backtracking	P1
17	Matchsticks to Square	Backtracking	P1
18	Beautiful Arrangement	Backtracking	P2
19	Letter Case Permutation	Backtracking	P1
20	Word Search II	Trie + backtracking	P1

14. Dynamic Programming

#	Question	Pattern	Priority
1	Climbing Stairs	1D DP	P0
2	House Robber	Pick/skip DP	P0
3	Coin Change	Unbounded DP	P0
4	Longest Increasing Subsequence	Sequence DP	P0
5	Word Break	DP + set	P0

#	Question	Pattern	Priority
6	Decode Ways	1D DP	P0
7	Unique Paths	2D DP	P1
8	Longest Common Subsequence	2D DP	P1
9	Maximum Product Subarray	DP state	P1
10	Partition Equal Subset Sum	0/1 knapsack	P1
11	Min Cost Climbing Stairs	1D DP	P1
12	House Robber II	Circular DP	P1
13	Palindromic Substrings	DP/center	P1
14	Longest Palindromic Substring	DP/center	P1
15	Target Sum	DP/counting	P1
16	Combination Sum IV	DP counting	P1
17	Edit Distance	2D DP	P2
18	Regular Expression Matching	2D DP	P2
19	Burst Balloons	Interval DP	P2
20	Best Time to Buy and Sell Stock with Cooldown	State DP	P1

15. Greedy

#	Question	Pattern	Priority
1	Jump Game	Greedy reach	P0
2	Jump Game II	Greedy levels	P1
3	Gas Station	Greedy reset	P1
4	Hand of Straights	Greedy + map	P1
5	Partition Labels	Greedy intervals	P1
6	Valid Parenthesis String	Greedy bounds	P1
7	Non-overlapping Intervals	Greedy intervals	P0
8	Minimum Number of Arrows to Burst Balloons	Greedy intervals	P1

#	Question	Pattern	Priority
9	Task Scheduler	Greedy + heap	P0
10	Queue Reconstruction by Height	Sort + greedy	P1
11	Candy	Two-pass greedy	P2
12	Lemonade Change	Greedy counting	P1
13	Assign Cookies	Greedy sort	P1
14	Boats to Save People	Greedy two pointers	P1
15	Maximum Subarray	Kadane/greedy	P0
16	Best Time to Buy and Sell Stock II	Greedy profit	P1
17	Merge Triplets to Form Target Triplet	Greedy check	P1
18	Valid Palindrome II	Greedy skip	P1
19	Minimum Deletions to Make Character Frequencies Unique	Greedy frequency	P1
20	Remove K Digits	Greedy stack	P1

16. Trie

#	Question	Pattern	Priority
1	Implement Trie	Trie basics	P0
2	Design Add and Search Words Data Structure	Trie + DFS	P1
3	Word Search II	Trie + backtracking	P1
4	Search Suggestions System	Trie/sorting	P1
5	Replace Words	Trie prefix	P1
6	Longest Word in Dictionary	Trie/sort	P1
7	Map Sum Pairs	Trie + prefix sum	P1
8	Prefix and Suffix Search	Trie/hash design	P2
9	Concatenated Words	Trie/DP	P2
10	Stream of Characters	Reverse trie	P2
11	Palindrome Pairs	Trie/string	P2
12	Maximum XOR of Two Numbers in an	Bit trie	P1

#	Question	Pattern	Priority
	Array		
13	Implement Magic Dictionary	Trie + DFS	P2
14	Design File System	Trie/tree	P1
15	Design Search Autocomplete System	Trie + heap	P1
16	Longest Common Prefix	Trie/string	P1
17	Word Squares	Trie + backtracking	P2
18	Camelcase Matching	Trie/string	P2
19	Short Encoding of Words	Trie/suffix	P2
20	Count Prefix and Suffix Pairs	Trie/string	P2

17. Bit Manipulation

#	Question	Pattern	Priority
1	Single Number	XOR	P1
2	Number of 1 Bits	Bit count	P1
3	Counting Bits	DP + bits	P1
4	Reverse Bits	Bit operations	P1
5	Missing Number	XOR/math	P1
6	Sum of Two Integers	Bit addition	P1
7	Bitwise AND of Numbers Range	Common prefix	P1
8	Power of Two	Bit trick	P1
9	Power of Four	Bit trick	P2
10	Single Number II	Bit counting	P2
11	Single Number III	XOR partition	P2
12	Subsets	Bitmask/backtracking	P1
13	Maximum Product of Word Lengths	Bitmask	P2
14	UTF-8 Validation	Bitmask parsing	P2
15	Hamming Distance	XOR	P1
16	Total Hamming Distance	Bit counting	P2
17	Find the Difference	XOR	P1

#	Question	Pattern	Priority
18	Complement of Base 10 Integer	Bitmask	P2
19	Gray Code	Bit generation	P2
20	Minimum Flips to Make a OR b Equal to c	Bit comparison	P2

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18. Absolute Minimum List If Time Becomes Very Tight

If you cannot complete every section, solve these first.

#	Question	Section
1	Two Sum	Arrays & Hashing
2	Group Anagrams	Arrays & Hashing
3	Product of Array Except Self	Arrays & Hashing
4	Longest Consecutive Sequence	Arrays & Hashing
5	Subarray Sum Equals K	Prefix Sum
6	3Sum	Two Pointers
7	Container With Most Water	Two Pointers
8	Longest Substring Without Repeating Characters	Sliding Window
9	Minimum Window Substring	Sliding Window
10	Search in Rotated Sorted Array	Binary Search
11	Koko Eating Bananas	Binary Search
12	Valid Parentheses	Stack
13	Daily Temperatures	Stack
14	Reverse Linked List	Linked List
15	Linked List Cycle	Linked List
16	Binary Tree Level Order Traversal	Trees
17	Validate Binary Search Tree	Trees
18	Kth Largest Element in an Array	Heap
19	Merge Intervals	Intervals
20	Number of Islands	Graphs
21	Course Schedule	Graphs
22	Climbing Stairs	DP

#	Question	Section
23	Coin Change	DP
24	Jump Game	Greedy
25	Implement Trie	Trie

---

19. Daily Rule

For every problem, write this before moving forward:

```
1. What is the brute force?
2. What is the optimized approach?
3. Which data structure is used?
4. Why this data structure?
5. Time complexity?
6. Space complexity?
7. Edge cases?
8. Dry run?
```

---

20. What to Skip Until This List Is Done

Skip these for now:

- Segment Tree
- Fenwick Tree
- Red-Black Tree implementation
- AVL Tree implementation
- Max Flow / Min Cut
- Heavy computational geometry
- Suffix Array
- Advanced competitive programming math
- Very hard DP before basic DP is strong

---

21. Final Advice

For your background as a web developer, your target is not to become a competitive programmer.

Your target is:

```
Recognize the pattern.
Pick the correct data structure.
Write clean TypeScript.
```

```
Explain time and space complexity.
Handle edge cases.
Dry run confidently.
```